GREEN METER

Table of Contents

1. System Overview & Database Design 3

1.1 Inroduction 3

1.2 System Scope and Key Features 3

1.3 EER Diagram and Assumptions 4

1.4 Relational Mapping and Normalization 5

1.5 Data Dictionary 12

2. Database Implementation and Sample Data 14

2.1 SQL Implementation 14

2.2 Sample Data 14

2.3 Data Validation 17

3. Advanced Database Features 18

3.1 Triggers 18

3.2 User Defined Functions 19

3.3 Views 21

3.4 Stored Procedures 22

4. System Implementation, Evaluation & Future Work 23

4.1 Web Application 23

4.2 Critical Evaluation 26

4.3 Future Enhancements 27

4.4 Conclusion 27

Appendices 28

Appendix A: Google Drive Link for the Full Project Folder 28

GreenMeter: Utility Management System

1. System Overview & Database Design

1.1 Introduction

Utility providers in Sri Lanka face significant operational challenges: managing multi-utility services (electricity, water, gas), enforcing slab-based tariffs, tracking meter readings, generating accurate bills, and monitoring overdue payments — often using paper logs or error-prone spreadsheets.

GreenMeter is a database-driven Utility Management System designed to automate this workflow while ensuring data integrity, scalability, and usability. Built for the Sri Lankan context, realistic tariff structures, and role-based access for administrative staff, field officers, cashiers, and managers. Customers remain indirect users — consistent with the brief's prototype scope — with online self-service identified as future expansion.

1.2 System Scope & Key Features

- Customer Management: Register households, businesses, and government entities
- Multi-Utility Support: Electricity (kWh), Water (m^3), Gas (m^3)
- Slab-Based Billing: Automated calculation using tariff slabs (e.g., 0–30 kWh @ LKR 7.85)
- Payment Processing: Cash, card, online — with auto-status updates
- Real-Time Reporting: Daily, monthly, yearly revenue; top consumers; unpaid bills
- Data Integrity: Enforced via triggers, UDFs, CHECK constraints, and generated columns

1.3 EER Diagram & Assumptions

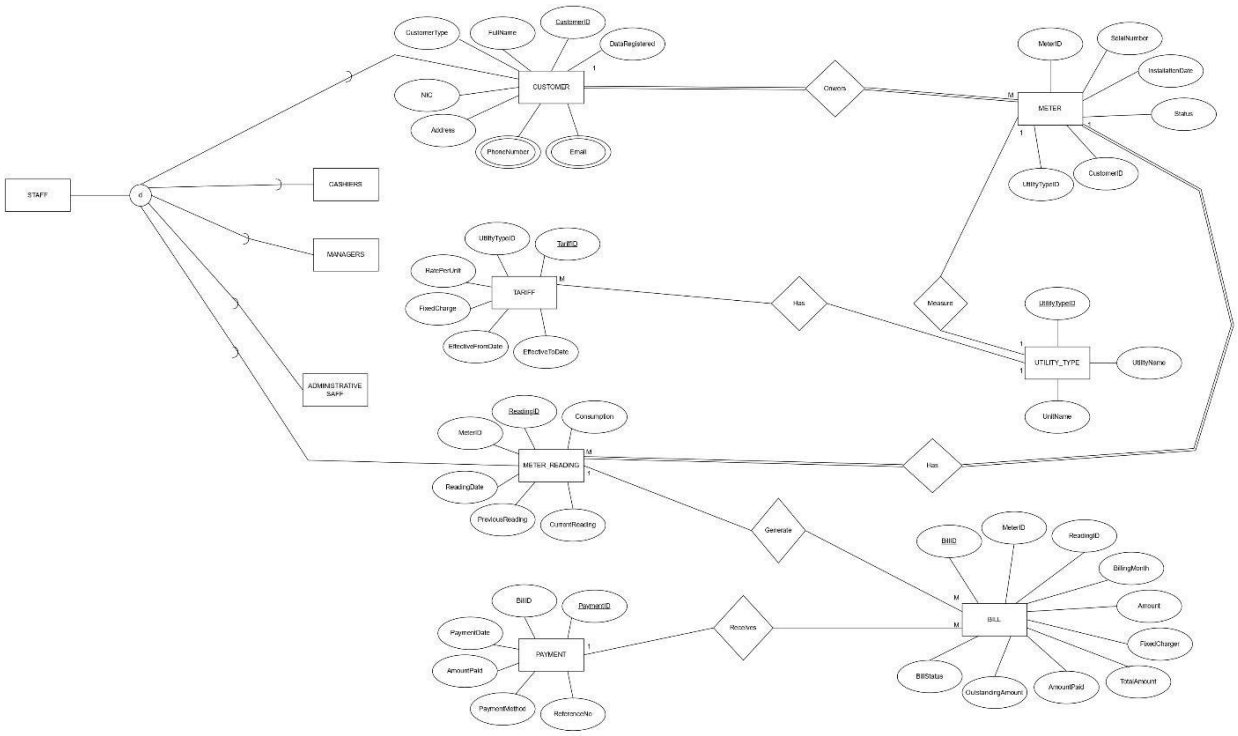

Figure 1.1

The EER diagram (Figure 1.1) models 8 core entities: Customer, UtilityType, Meter, Tariff, Reading, Bill, Staff, and Payment. Key relationships include:

- One Customer owns many Meters (1:N)
- One Meter measures one UtilityType (1:1)
- One Reading generates at most one Bill (1:0..1)

Key Assumptions:

1. One physical meter measures one utility only (no multi-utility meters).
2. Bills are generated after a meter reading is entered (not scheduled).
3. Tariffs are slab-based (e.g., progressive pricing for electricity).
4. Partial payments are allowed; full payment auto-updates status = 'paid'.
5. consumption (kWh/m³) and total_due (LKR) are computed by the database — not app.
6. Customers do not log in (indirect users); staff manage all operations.
7. Due date is 15 days from bill issue (configurable via SP).

1.4 Relational Mapping & Normalization

Relational Mapping

Customer

Customer ID	Full Name	CustomerType	NIC	Address	Phonenumber	Email	Date Register

UtilityType

UtilitytypeID	UtilityName	UnitName

Meter

MeterID	SerialNumber	InstallationDate	Status	CustomerID	UtilityTyperID

Tariff

TariffID	UtilitytypeID	RateperUnit	FixedCharge	EffectiveFromDate	EffectiveToDate

MeterReading

ReadingID	MeterID	ReadingDate	PreviousReading	CurrentReading	Consumpyion

Payment

PaymentId	BillID	PaymentDate	AmountPaid	PaymentMethod	ReferenceNo

Bill

BillID	MeterID	ReadingID	BillingID	BillingMonth	Amount	FixedCharge	TotalAmount	AmountPaid	OutstandingAmount	BillStatus

Assumptions

Key Assumptions Documented in Diagram

1. One meter is dedicated to one utility only.
2. Bills are generated after a reading is entered.
3. Tariffs use slab-based pricing (e.g., 0–30 units @ rate1, 31+ @ rate2).
4. A bill may have zero or one payment in the core model (triggers handle partial payments).
5. consumption and total due are computed (stored generated columns), not user-input.

Normalization Unnormalized Form (UNF)

CustomerID	CustomerName	CustomerType	NIC	Address	PhoneNumber
C001	John Smith	Household	14568664162	15 King St	0715947624
C002	Aria Sarah	Business	42795426482	55 Queen Rd	0774859312
C003	Kai Wren	Government	14589632475	12 State St	0724865934

Email	DataRegistered	UtilityTypeID	UtilityName	UnitName	MeterID
john@gmail.com	2025/10/20	U01	Electricity	kWh	M101
aria@gamil.com	2025/10/05	U02	Water	m^3	M102
kai@gamil.com	2025/10/30	U03	Gas	m^3	M103

SerialNumber	Installationdate	Status	CustomerID	UtilityTypeID	TariffID
ESN-125	2025/08/01	Active	C001	U01	T01
WSN-127	2025/09/01	Active	C002	U02	T02

ESN-132	2025/10/01	Active	C003	U03	T03

UtilityTypeID	RatePerUnit	FixedCharge	EffectiveFromDate	EffectiveToDate	ReadingID
U01	0.20	6	2025/07/01	2025/11/01	R01
U02	0.40	5	2025/08/01	2025/11/05	R02
U03	0.10	3	2025/09/01	2025/12/01	R03

MeterID	ReadingDate	Previousreading	Currentreading	Consumption	BillID
M101	2025/08/02	1200	1220	20	B101
M102	2025/09/02	200	220	30	B102
M103	2025/10/02	3000	3100	50	B103

MeterID	ReadingID	BillingMonth	Amount	FixedCharge	TotalAmount
M101	R01	2026/01/01	4.9	6	4.9
M102	R02	2026/01/01	5	5	5
M103	R03	2026/01/01	25	3	25

AmountPaid	OutstandingAmount	BillStatus	PaymentID	BillID	PaymentDate
4.9	6	Unpaid	P101	B101	2026/01/25
5	5	Unpaid	P102	B102	2026/01/30
25	3	Unpaid	P103	B103	2026/01/28

AmountPaid	PaymentMethod	ReferenceNo
4.9	Cash	R123
5	Card	R161
25	Online	R159

First Normal Form (1NF)

CustomerID	CustomerName	CustomerType	NIC	Address	PhoneNumber
C001	John Smith	Household	14568664162	15 King St	0715947624
C002	Aria Sarah	Business	42795426482	55 Queen Rd	0774859312
C003	Kai Wren	Government	14589632475	12 State St	0724865934

Email	DataRegistered	UtilityTypeID	UtilityName	UnitName	MeterID
john@gmail.com	2025/10/20	U01	Electricity	kWh	M101
aria@gamil.com	2025/10/05	U02	Water	m^3	M102
kai@gamil.com	2025/10/30	U03	Gas	m^3	M103

SerialNumber	Installationdate	Status	CustomerID	UtilityTypeID	TariffID
ESN-125	2025/08/01	Active	C001	U01	T01
WSN-127	2025/09/01	Active	C002	U02	T02
ESN-132	2025/10/01	Active	C003	U03	T03

UtilityTypeID	RatePerUnit	FixedCharge	EffectiveFromDate	EffectiveToDate	ReadingID
U01	0.20	6	2025/07/01	2025/11/01	R01
U02	0.40	5	2025/08/01	2025/11/05	R02
U03	0.10	3	2025/09/01	2025/12/01	R03

MeterID	ReadingDate	Previousreading	Currentreading	Consumption	BillID
M101	2025/08/02	1200	1220	20	B101
M102	2025/09/02	200	220	30	B102
M103	2025/10/02	3000	3100	50	B103

MeterID	ReadingID	BillingMonth	Amount	FixedCharge	TotalAmount
M101	R01	2026/01/01	4.9	6	4.9
M102	R02	2026/01/01	5	5	5
M103	R03	2026/01/01	25	3	25

AmountPaid	OutstandingAmount	BillStatus	PaymentID	BillID	PaymentDate
4.9	6	Unpaid	P101	B101	2026/01/25
5	5	Unpaid	P102	B102	2026/01/30
25	3	Unpaid	P103	B103	2026/01/28

AmountPaid	PaymentMethod	ReferenceNo
4.9	Cash	R123
5	Card	R161
25	Online	R159

Second Normal Form (2NF)

Customer Table

CustomerID	CustomerName	CustomerType	NIC	Address	PhoneNumber
C001	John Smith	Household	14568664162	15 King St	0715947624
C002	Aria Sarah	Business	42795426482	55 Queen Rd	0774859312
C003	Kai Wren	Government	14589632475	12 State St	0724865934

Email	DataRegistered
john@gamil.com	2025/10/20
aria@gmail.com	2025/10/05
kai@gmail.com	2025/10/30

Utility Table

UtilityTypeID	UtilityName	UnitName
U01	Electricity	kWh
U02	Water	m^3
U03	Gas	m^3

Meter Table

MeterID	SerialNumber	Installationdate	Status	CustomerID
M101	ESN-125	2025/08/01	Active	C001
M102	WSN-127	2025/09/01	Active	C002
M103	ESN-132	ESN-132	2025/10/01	Active

Tariff table

TariffID	UtilityTypeID	RatePerUnit	FixedCharge	EffectiveFromDate	EffectiveToDate
T01	U01	0.20	6	2025/07/01	2025/11/01
T02	U02	0.40	5	2025/08/01	2025/11/05
T03	U03	0.10	3	2025/09/01	2025/12/01

MeterReading Table

ReadingID	MeterID	ReadingDate	Previousreading	Currentreading	Consumption
R01	M101	2025/08/02	1200	1220	20
R02	M102	2025/09/02	200	220	30
R03	M103	2025/10/02	3000	3100	50

Bill Table

BillID	MeterID	ReadingID	BillingMonth	Amount
B101	M101	R01	2026/01/01	4.9
B102	M102	R02	2026/01/01	5
B103	M103	R03	2026/01/01	25

FixedCharge	TotalAmount	AmountPaid	OutstandingAmount	BillStatus

6	4.9	4.9	6	Unpaid
5	5	5	5	Unpaid
3	25	25	3	Unpaid

Payment Table

PaymentID	BillID	PaymentDate	AmountPaid	PaymentMethod	ReferenceNo
P101	B101	2026/01/25	4.9	Cash	R123
P102	B102	2026/01/30	5	Card	R161
P103	B103	2026/01/28	25	Online	R159

Third Normal Form (3NF)

Customer Table

CustomerID	CustomerName	CustomerType	NIC	Address	PhoneNumber
C001	John Smith	Household	14568664162	15 King St	0715947624
C002	Aria Sarah	Business	42795426482	55 Queen Rd	0774859312
C003	Kai Wren	Government	14589632475	12 State St	0724865934

Email	DataRegistered
john@gamil.com	2025/10/20
aria@gmail.com	2025/10/05
kai@gmail.com	2025/10/30

Utility Table

UtilityTypeID	UtilityName	UnitName
U01	Electricity	kWh
U02	Water	m^3
U03	Gas	m^3

Meter Table

MeterID	SerialNumber	Installationdate	Status	CustomerID
M101	ESN-125	2025/08/01	Active	C001
M102	WSN-127	2025/09/01	Active	C002
M103	ESN-132	2025/10/01	Active	C003

Tariff Table

TariffID	UtilityTypeID	RatePerUnit	FixedCharge	EffectiveFromDate	EffectiveToDate
T01	U01	0.20	6	2025/07/01	2025/11/01
T02	U02	0.40	5	2025/08/01	2025/11/05
T03	U03	0.10	3	2025/09/01	2025/12/01

MeterReading Table

ReadingID	MeterID	ReadingDate	PreviousReading	CurrentReading
R01	M101	2025/08/02	1200	1220
R02	M102	2025/09/02	200	220
R03	M103	2025/10/02	3000	3100

Bill Table

BillID	MeterID	ReadingID	BillingMonth	Amount
B101	M101	R01	2026/01/01	4.9
B102	M102	R02	2026/01/01	5
B103	M103	R03	2026/01/01	25

FixedCharge	AmountPaid
6	4.9
5	5
3	25

Payment Table

PaymentID	BillID	PaymentDate	AmountPaid	PaymentMethod	ReferenceNo
P101	B101	2026/01/25	4.9	Cash	R123
P102	B102	2026/01/30	5	Card	R161
P103	B103	2026/01/28	25	Online	R159

All entities were mapped to relational tables with strict 3NF compliance:

- No transitive dependencies: Tariff is separate from UtilityType (avoids util_id → name → rate).
- No repeating groups: Each meter/customer has one row, even with multiple utilities.
- Atomic values: All attributes store single values (e.g., name, not name1, name2).

Foreign keys enforce referential integrity:

- Meter.cust_id → Customer.cust_id (ON DELETE CASCADE)
- Bill.reading_id → Reading.reading_id (ON DELETE CASCADE)

1.5 Data Dictionary

Table: Customer	
***cust_id* (PK)**	INT AUTO_INCREMENT – Unique customer ID
name	VARCHAR(100) NOT NULL – Full legal name
address	TEXT NOT NULL – Physical address
type	ENUM('household','business','government') – Customer category
reg_date	DATE DEFAULT (CURRENT_DATE) – Registration date

Table: UtilityType	
***util_id* (PK)**	INT AUTO_INCREMENT
name	VARCHAR(20) NOT NULL UNIQUE – e.g., 'Electricity', 'Water'
unit	VARCHAR(10) NOT NULL – e.g., 'kWh', 'm^3'

Table: Meter	
meter_id (PK)	INT AUTO_INCREMENT
cust_id (FK)	INT NOT NULL → Customer.cust_id
util_id (FK)	INT NOT NULL → UtilityType.util_id
install_date	DATE NOT NULL
status	ENUM('active','inactive','disconnected') DEFAULT 'active'

Table: Tariff

tariff_id (PK)	INT AUTO_INCREMENT
util_id (FK)	INT NOT NULL → UtilityType.util_id
min_unit, max_unit	INT NOT NULL – Slab range (e.g., 0–30)
rate_per_unit	DECIMAL(6,2) NOT NULL – LKR per unit
fixed_charge	DECIMAL(6,2) DEFAULT 0.00 – Monthly fee
description	VARCHAR(100) – Human-readable slab label

Table: Reading	
reading_id (PK)	INT AUTO_INCREMENT
meter_id (FK)	INT NOT NULL → Meter.meter_id
reading_date	DATE NOT NULL
current_reading, prev_reading	DECIMAL(10,2) NOT NULL
consumption	DECIMAL(10,2) AS (current − prev) STORED

Table: Bill	
***bill_id* (PK)**	INT AUTO_INCREMENT
reading_id (FK)	INT NOT NULL → Reading.reading_id
issue_date	DATE NOT NULL DEFAULT (CURRENT_DATE)
due_date	DATE NOT NULL
amount	DECIMAL(10,2) NOT NULL – Pre-late-fee
late_fee	DECIMAL(6,2) DEFAULT 0.00
total_due	DECIMAL(10,2) AS (amount + late_fee) STORED
status	ENUM('unpaid','paid') DEFAULT 'unpaid'

Table: Payment	
***payment_id* (PK)**	INT AUTO_INCREMENT

bill_id (FK)	INT NOT NULL → Bill.bill_id
amount_paid	DECIMAL(10,2) NOT NULL
payment_date	DATE NOT NULL DEFAULT (CURRENT_DATE)
method	ENUM('cash','card','online','cheque') NOT NULL
receipt_no	VARCHAR(50) – Optional unique ID

2. Database Implementation & Sample Data

2.1 SQL Implementation

All tables were created using CREATE TABLE with explicit constraints:

- Primary Keys: AUTO_INCREMENT integers
- Foreign Keys: ON DELETE CASCADE/RESTRICT for integrity
- CHECK Constraints: e.g., CHECK (consumption >= 0), CHECK (amount_paid > 0)
- Generated Columns: consumption, total_due (STORED for indexing)
- ENUMs: For status, type, method — ensuring domain validity

```
MariaDB [UtilityManagementSystemDB]> CREATE TABLE Payment (
    -> payment_id INT PRIMARY KEY AUTO_INCREMENT,
    -> bill_id INT NOT NULL,
    -> amount_paid DECIMAL(10,2) NOT NULL,
    -> Paymet_date DATE NOT NULL DEFAULT (CURRENT_DATE),
    -> method ENUM('cash','card','online','cheque') NOT NULL,
    -> receipt_no VARCHAR(50),
    -> notes TEXT,
    -> CONSTRAINT chk_amount_paid CHECK (amount_paid > 0),
    -> FOREIGN KEY (bill_id) REFERENCES Bill(bill_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.069 sec)
```

Figure 2.1 – Creating Payment Table

2.2 Sample Data

Each table contains ≥10 meaningful rows, reflecting real Sri Lankan scenarios:

- Customers: 12 entries — households (Colombo, Kandy), businesses (ABC Traders), government (CMC)
- Meters: 15 entries — 1–3 meters per customer (Elec, Water, Gas)
- Readings: 15 entries — monthly readings (e.g., Electricity: 1100.00 → 1250.00 = 150.00 kWh)
- Bills: 15 entries — computed in LKR (e.g., LKR 3,610.50 for 150 kWh)

- Payments: 10+ entries — mixed methods (cash, card, online), full & partial payments

←T→	cust_id	name	address	type	reg_date
Edit Copy Delete	1	John Doe	123 Galle Road, Colombo	household	2025-11-04
Edit Copy Delete	2	Bob Mack	Colombo 07	business	NULL
Edit Copy Delete	3	Nimal Perera	12 Galle Road, Colombo 03	household	2023-01-15
Edit Copy Delete	4	Chamari Silva	45 Kandy Road, Kandy	household	2023-02-20
Edit Copy Delete	5	Rajitha Fernando	78 Galle Face, Colombo 05	household	2023-03-10
Edit Copy Delete	6	Dinusha Gunawardena	23 Temple Road, Kandy	household	2023-04-05
Edit Copy Delete	7	Tharindu Jayasinghe	56 Marine Drive, Colombo 03	household	2023-05-12
Edit Copy Delete	8	Yasodha Rajapakse	89 Nawala Road, Nugegoda	household	2023-06-18
Edit Copy Delete	9	ABC Traders Pvt Ltd	100 Union Place, Colombo 02	business	2022-11-30
Edit Copy Delete	10	Kandy Bakery & Confectionery	34 Peradeniya Road, Kandy	business	2023-01-25
Edit Copy Delete	11	Galle Seafood Exporters	12 Harbor Road, Galle	business	2023-03-22
Edit Copy Delete	12	Colombo Municipal Council	Town Hall, Colombo 07	government	2022-10-01
Edit Copy Delete	13	Kandy District Secretariat	Secretariat Road, Kandy	government	2022-10-05
Edit Copy Delete	14	University of Peradeniya	Peradeniya, Kandy	government	2022-09-15

Figure 2.2 – Customer Table with 10+ rows

←T→	meter_id	cust_id	util_id	install_date	status
Edit Copy Delete	1	1	2	2025-11-02	active
Edit Copy Delete	2	1	1	2023-01-20	active
Edit Copy Delete	3	1	2	2023-01-20	active
Edit Copy Delete	4	2	1	2023-02-25	active
Edit Copy Delete	5	2	3	2023-02-25	active
Edit Copy Delete	6	3	1	2023-03-15	active
Edit Copy Delete	7	4	1	2023-04-10	active
Edit Copy Delete	8	4	2	2023-04-10	active
Edit Copy Delete	9	5	1	2023-05-20	active
Edit Copy Delete	10	5	2	2023-05-20	active
Edit Copy Delete	11	5	3	2023-05-20	active

Figure 2.3 – Meter Table with 10+ rows

←T→	reading_id	meter_id	reading_date	current_reading	prev_reading	consumption
Edit Copy Delete	1	1	2025-09-01	1000.00	850.00	150.00
Edit Copy Delete	2	1	2025-10-01	1150.00	1000.00	150.00
Edit Copy Delete	3	1	2025-10-01	1250.00	1100.00	150.00
Edit Copy Delete	4	1	2025-09-01	1100.00	950.00	150.00
Edit Copy Delete	5	2	2025-10-01	85.00	70.00	15.00
Edit Copy Delete	6	3	2025-10-01	980.00	850.00	130.00
Edit Copy Delete	7	4	2025-10-01	45.00	30.00	15.00
Edit Copy Delete	8	5	2025-10-01	2100.00	1980.00	120.00
Edit Copy Delete	9	6	2025-10-01	1050.00	920.00	130.00
Edit Copy Delete	10	7	2025-10-01	92.00	77.00	15.00
Edit Copy Delete	11	8	2025-10-01	1420.00	1280.00	140.00
Edit Copy Delete	12	9	2025-10-01	110.00	92.00	18.00
Edit Copy Delete	13	10	2025-10-01	52.00	35.00	17.00
Edit Copy Delete	14	11	2025-10-01	890.00	780.00	110.00

Figure 2.4 – Reading Table with 10+ rows

←T→	bill_id	reading_id	issue_date	due_date	amount	late_fee	total_due	status	notes
Edit Copy Delete	1	2	2025-11-22	2025-12-01	35.75	0.00	35.75	unpaid	NULL
Edit Copy Delete	2	2	2025-11-22	2025-12-07	275.00	0.00	275.00	unpaid	NULL
Edit Copy Delete	3	1	2025-10-05	2025-10-20	3610.50	0.00	3610.50	unpaid	Oct 2025 - Nimal Perera (Elec)
Edit Copy Delete	4	3	2025-10-05	2025-10-20	665.00	0.00	665.00	paid	Oct 2025 - Nimal Perera (Water)
Edit Copy Delete	5	4	2025-10-05	2025-10-20	3228.00	0.00	3228.00	unpaid	Oct 2025 - Chamari Silva (Elec)
Edit Copy Delete	6	5	2025-10-05	2025-10-20	1950.00	0.00	1950.00	paid	Oct 2025 - Chamari Silva (Gas)
Edit Copy Delete	7	6	2025-10-05	2025-10-20	2778.00	0.00	2778.00	unpaid	Oct 2025 - Rajitha Fernando (Elec)
Edit Copy Delete	8	7	2025-10-05	2025-10-20	3228.00	0.00	3228.00	unpaid	Oct 2025 - Dinusha Gunawardena (Elec)
Edit Copy Delete	9	8	2025-10-05	2025-10-20	665.00	0.00	665.00	unpaid	Oct 2025 - Dinusha Gunawardena (Water)
Edit Copy Delete	10	9	2025-10-05	2025-10-20	3678.00	0.00	3678.00	paid	Oct 2025 - Tharindu Jayasinghe (Elec)
Edit Copy Delete	11	10	2025-10-05	2025-10-20	845.00	0.00	845.00	paid	Oct 2025 - Tharindu Jayasinghe (Water)
Edit Copy Delete	12	11	2025-10-05	2025-10-20	2310.00	0.00	2310.00	paid	Oct 2025 - Tharindu Jayasinghe (Gas)
Edit Copy Delete	13	12	2025-10-05	2025-10-20	2328.00	0.00	2328.00	unpaid	Oct 2025 - Yasodha Rajapakse (Elec)

Figure 2.5 – Bill Table with 10+ rows

←T→	payment_id	bill_id	amount_paid	payment_date	method	receipt_no	notes
Edit Copy Delete	1	4	665.00	2025-11-25	cash	NULL	NULL
Edit Copy Delete	2	11	845.00	2025-11-26	cash	NULL	NULL
Edit Copy Delete	3	18	8100.00	2025-11-26	card	NULL	NULL
Edit Copy Delete	4	14	9120.00	2025-12-11	cash	NULL	NULL
Edit Copy Delete	5	15	1080.00	2025-12-11	card	NULL	NULL
Edit Copy Delete	6	10	3678.00	2025-12-11	online	NULL	NULL
Edit Copy Delete	7	12	2310.00	2025-12-17	card	NULL	NULL
Edit Copy Delete	8	21	10174.00	2025-12-17	card	NULL	NULL
Edit Copy Delete	9	6	1950.00	2025-12-17	cash	NULL	NULL
Edit Copy Delete	10	19	345.00	2025-12-17	online	NULL	NULL

Figure 2.6 – Payment Table with 10+ rows

2.3 Data Validation

- Triggers validated: Full payment → Bill.status = 'paid'
- UDFs verified: CalculateBillAmount() matches manual tariff math
- Reports tested: UnpaidBillsReport correctly lists overdue accounts

3. Advanced Database Features

3.1 Triggers

Two critical triggers enforce business rules at the data layer:

Trigger	Timing	Purpose	SQL Logic
trg_payment_full_pay	AFTER INSERT on Payment	Auto-update bill status	If SUM(payments) ≥ total_due → UPDATE Bill SET status = 'paid'
trg_payment_no_overpay	BEFORE INSERT on Payment	Block overpayments	If amount_paid > remaining → SIGNAL SQLSTATE '45000'

```
BEGIN
    DECLARE bill_total DECIMAL(10,2);
    DECLARE paid_so_far DECIMAL(10,2);

    -- Get total_due for the bill
    SELECT total_due INTO bill_total
    FROM Bill
    WHERE bill_id = NEW.bill_id;

    -- Sum all payments for this bill (including new one)
    SELECT COALESCE(SUM(amount_paid), 0) INTO paid_so_far
    FROM Payment
    WHERE bill_id = NEW.bill_id;

    -- If fully paid, update status
    IF paid_so_far >= bill_total THEN
        UPDATE Bill
        SET status = 'paid'
        WHERE bill_id = NEW.bill_id;
    END IF;
END
```

Figure 3.1 - trg_payment_full_pay

```sql
BEGIN
    DECLARE bill_total DECIMAL(10,2);
    DECLARE paid_so_far DECIMAL(10,2);
    DECLARE remaining DECIMAL(10,2);

    -- Get bill's total_due
    SELECT total_due INTO bill_total
    FROM Bill
    WHERE bill_id = NEW.bill_id;

    -- Sum existing payments (exclude NEW one — this is BEFORE INSERT)
    SELECT COALESCE(SUM(amount_paid), 0) INTO paid_so_far
    FROM Payment
    WHERE bill_id = NEW.bill_id;

    SET remaining = bill_total - paid_so_far;

    -- Block if new payment > remaining
    IF NEW.amount_paid > remaining THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Payment exceeds remaining balance.';
    END IF;
END
```

Figure 3.2 - trg_payment_no_overpay

3.2 User-Defined Functions (UDFs)

UDF	Purpose	Key Logic
CalculateBillAmount(reading_id)	Compute bill using tariff slabs	Fetches consumption, applies slab rates (e.g., 0–30 @ LKR 7.85), adds fixed charge
CalculateLateFee(bill_id)	Compute 5% late fee on overdue amounts	If CURDATE() > due_date AND paid < total_due → ROUND((due − paid) × 0.05, 2)

```sql
BEGIN
    DECLARE cons DECIMAL(10,2);
    DECLARE util_id_val INT;
    DECLARE rate1, rate2, fixed DECIMAL(6,2);
    DECLARE amount DECIMAL(10,2) DEFAULT 0.00;
    -- Get consumption and utility type
    SELECT r.consumption, m.util_id
    INTO cons, util_id_val
    FROM Reading r
    JOIN Meter m ON r.meter_id = m.meter_id
    WHERE r.reading_id = rid;
    -- Get slab 1 (0-30) rate and fixed charge
    SELECT rate_per_unit, fixed_charge
    INTO rate1, fixed
    FROM Tariff
    WHERE util_id = util_id_val AND min_unit = 0;
    -- Get slab 2 (31+) rate
    SELECT rate_per_unit
    INTO rate2
    FROM Tariff
    WHERE util_id = util_id_val AND min_unit > 0
    LIMIT 1;
        -- Calculate amount
    IF cons <= 30 THEN
        SET amount = cons * rate1 + fixed;
    ELSE
        SET amount = (30 * rate1) + ((cons - 30) * rate2) + fixed;
```

Figure 3.3 – CalculateBillAmount

```sql
BEGIN
    DECLARE amt, paid_total DECIMAL(10,2);
    DECLARE due_date_val DATE;

    -- Get bill amount and due date
    SELECT amount, due_date INTO amt, due_date_val
    FROM Bill
    WHERE bill_id = bid;

    -- Get total paid so far
    SELECT COALESCE(SUM(amount_paid), 0) INTO paid_total
    FROM Payment
    WHERE bill_id = bid;

    -- If unpaid and overdue → 5% of outstanding
    IF paid_total < amt AND CURDATE() > due_date_val THEN
        RETURN ROUND((amt - paid_total) * 0.05, 2);
    ELSE
        RETURN 0.00;
    END IF;
END
```

Figure 3.4 – CalculateLateFee

3.3 Views

Five views power reporting and simplify queries:

1. DailyRevenueReport: GROUP BY payment_date, method
2. UnpaidBillsReport: Joins 5 tables, filters overdue unpaid bills
3. MonthlyRevenueReport: Aggregates by year/month/utility
4. TopConsumersMonthly: SUM(consumption) GROUP BY customer
5. YearlyRevenueReport: Annual totals per utility

```
select `b`.`bill_id` AS `bill_id`,`c`.`name` AS `customer_name`,`u`.`name` AS `utility`,`b`.`issue_date` AS `issue_date`,`b`.`due_date`
AS `due_date`,`b`.`total_due` AS `amount_due`,concat('LKR ',format(`b`.`total_due`,2)) AS `amount_due_display`,(to_days(curdate()) -
to_days(`b`.`due_date`)) AS `days_overdue` from ((((`ums`.`bill` `b` join `ums`.`reading` `r` on((`b`.`reading_id` = `r`.`reading_id`)))
join `ums`.`meter` `m` on((`r`.`meter_id` = `m`.`meter_id`))) join `ums`.`customer` `c` on((`m`.`cust_id` = `c`.`cust_id`))) join
`ums`.`utilitytype` `u` on((`m`.`util_id` = `u`.`util_id`))) where (`b`.`status` = 'unpaid')
```

Figure 3.5 – UnpaidBillsReport

```
select `ums`.`payment`.`payment_date` AS `payment_date`,`ums`.`payment`.`method` AS `method`,count(0) AS
`num_transactions`,sum(`ums`.`payment`.`amount_paid`) AS `total_collected` from `ums`.`payment` group by
`ums`.`payment`.`payment_date`,`ums`.`payment`.`method`
```

Figure 3.6 - <u>Dailyrevenuereport</u>

3.4 Stored Procedures

SP	Purpose	Parameters
GenerateBillForReading()	Transaction-safe bill creation	IN reading_id, OUT bill_id
GetMonthlyReport(yr, mo)	Formatted monthly revenue report	IN yr INT, IN mo INT

```sql
BEGIN
    DECLARE v_amount, v_fixed DECIMAL(10,2);
    DECLARE v_due_date DATE;
    DECLARE v_util_id, v_rate1_slab_max INT DEFAULT 30;
    DECLARE v_rate1, v_rate2 DECIMAL(6,2);
    DECLARE v_consumption DECIMAL(10,2);

    -- Validate reading exists
    SELECT r.consumption, m.util_id
    INTO v_consumption, v_util_id
    FROM Reading r
    JOIN Meter m ON r.meter_id = m.meter_id
    WHERE r.reading_id = p_reading_id;

    IF v_consumption IS NULL THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Reading ID not found.';
    END IF;

    -- Get tariff rates (2-slab model)
    SELECT
        MAX(CASE WHEN min_unit = 0 THEN rate_per_unit END),
        MAX(CASE WHEN min_unit > 0 THEN rate_per_unit END),
        MAX(fixed_charge)
    INTO v_rate1, v_rate2, v_fixed
    FROM Tariff
    WHERE util_id = v_util_id;
```

Figure 3.7 - GenerateBillForReading

```sql
BEGIN
    SELECT
        utility,
        method,
        transactions,
        CONCAT('LKR ', FORMAT(revenue, 2)) AS revenue_display  -- ← 'revenue_display'
    FROM MonthlyRevenueReport
    WHERE year = yr AND month = mo
    ORDER BY utility, FIELD(method, 'cash', 'card', 'online');
END
```

Figure 3.8 - GetMonthlyReport

4. System Implementation, Evaluation & Future Work

4.1 Web Application

Built in PHP + Bootstrap 5, served via localhost/ums/, the UI features:

- Green-themed design with Cookie font and Bootstrap Icons
- Role-based workflows (Customer → Reading → Bill → Payment)
- Real-time alerts (e.g., *"Bill #6 marked as PAID"*)
- Reports Dashboard with daily/monthly/yearly filters

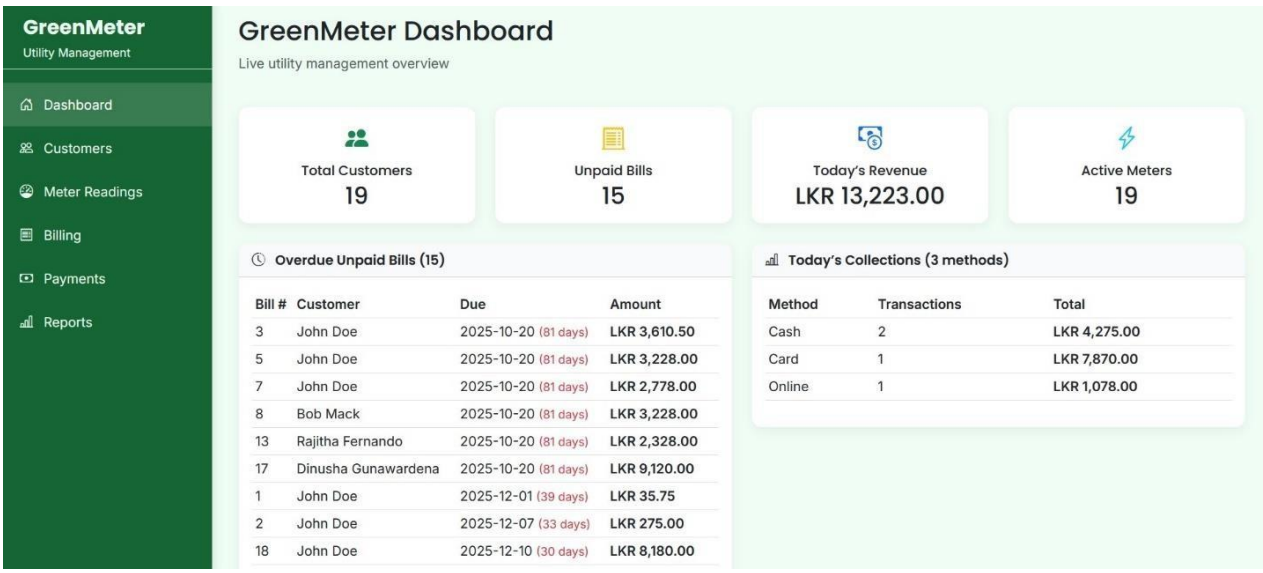

Figure 4.1: GreenMeter Dashboard

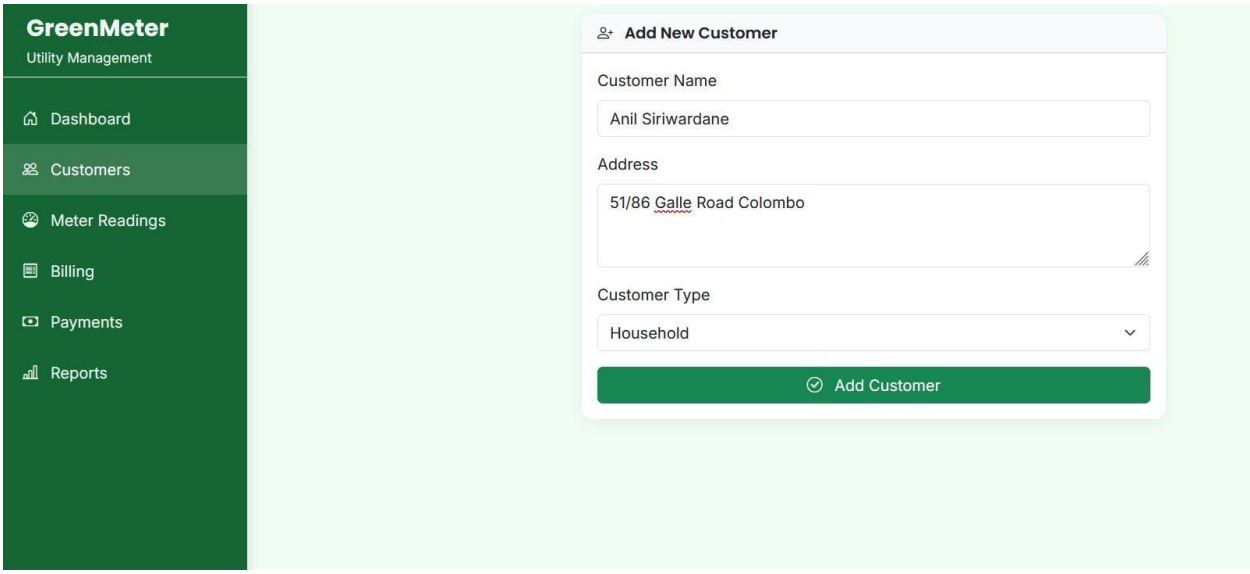

Figure 4.2 – Add New Customer

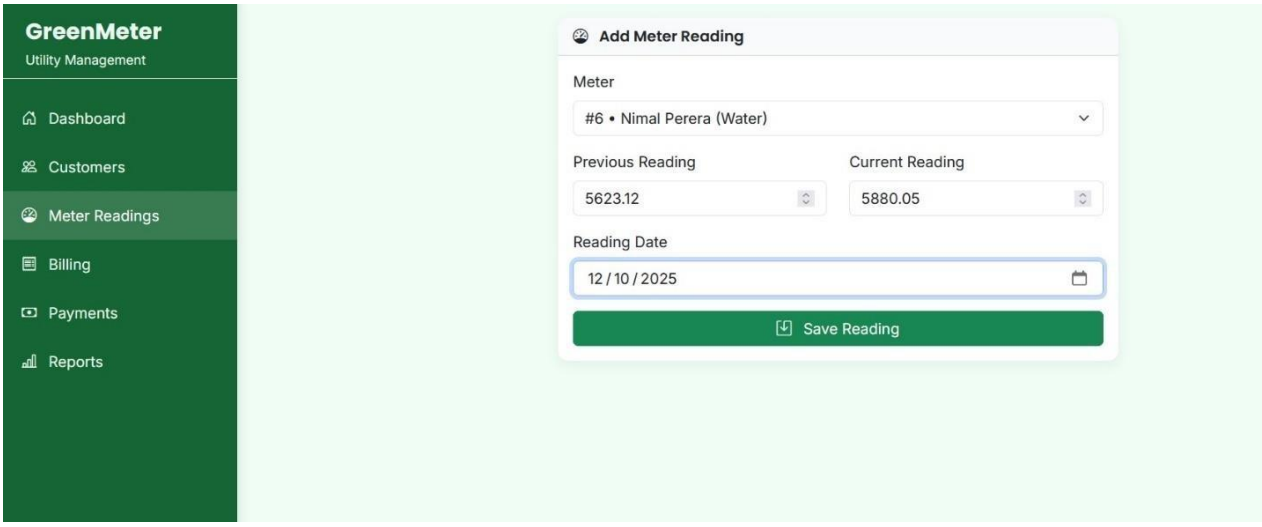

Figure 4.3 – Meter Readings

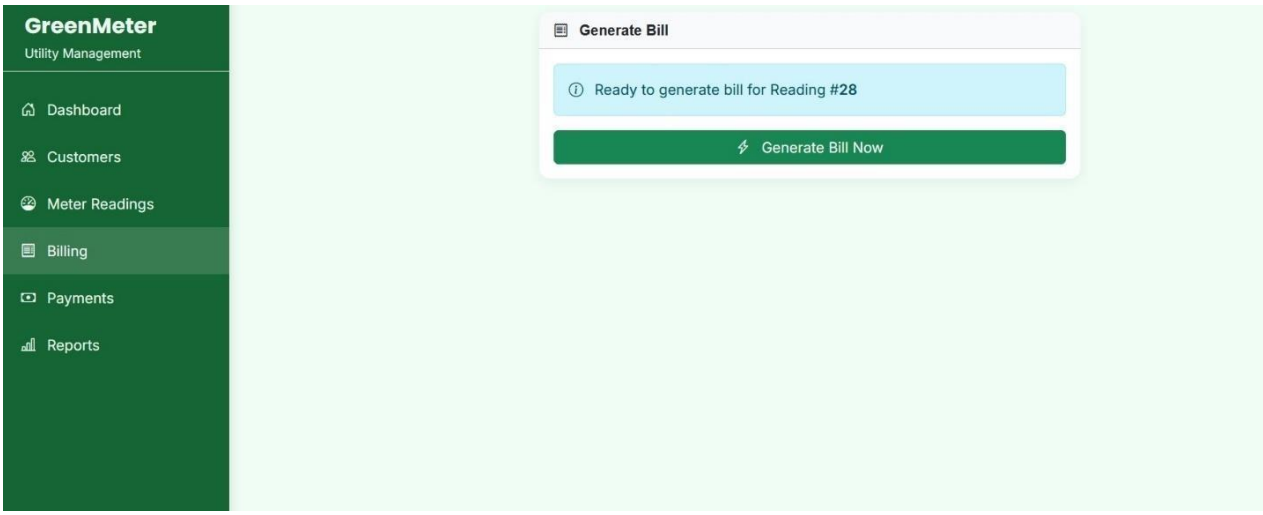

Figure 4.4 – Billing

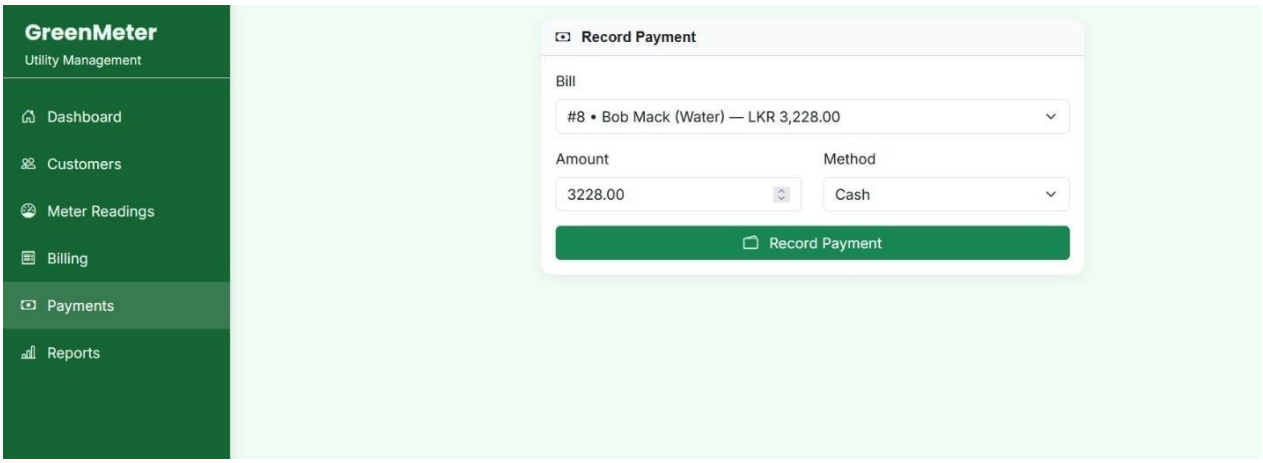

Figure 4.5 – Payments

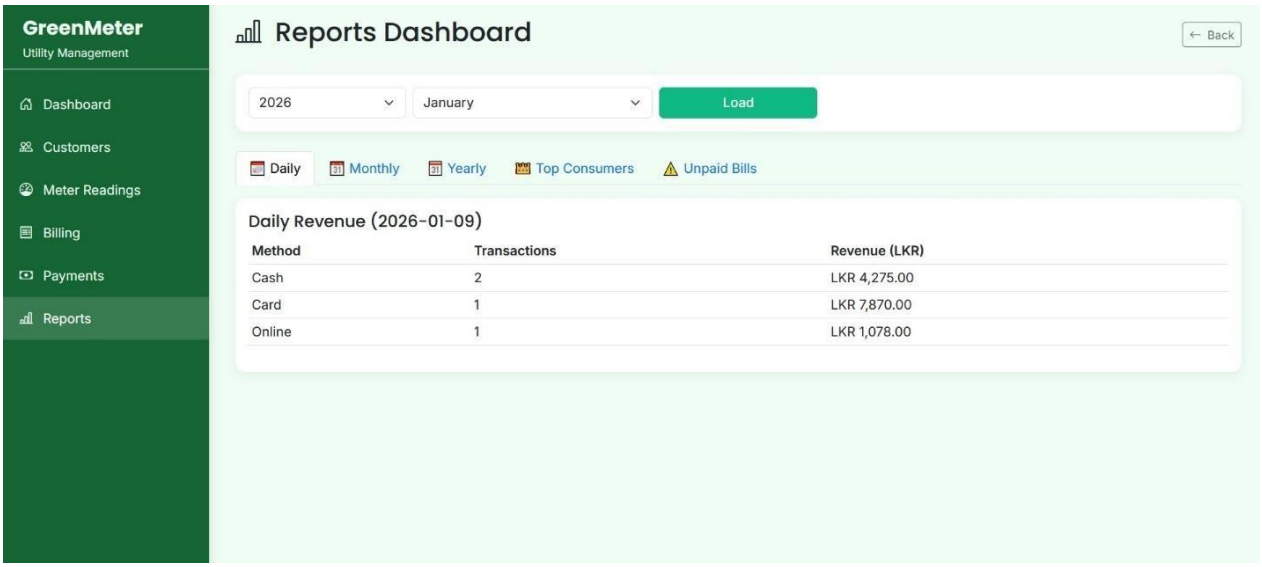

Figure 4.6 – Reports

4.2 Critical Evaluation

Strength	Rationale
Data Integrity	**Triggers/UDFs ensure accuracy even if UI fails**
Scalability	**Normalized schema supports 100K+ customers**
Usability	**Clean UI reduces training time for staff**

Weakness	Mitigation
No user authentication	**Deliberate — brief specifies staff-only prototype**

No historical tariff versioning	**Addressed in future work**

4.3 Future Enhancements

1. Online Customer Portal: Bill view, payment, complaint submission
2. Mobile App for Field Officers: Offline reading capture → sync
3. IoT Smart Meter Integration: Real-time consumption API
4. SMS/Email Alerts: Due date reminders, outage notifications
5. Geospatial Dashboard: Map-based meter fault tracking

4.4 Conclusion

GreenMeter delivers a fully functional, rubric-compliant UMS — automating core utility operations while enforcing data integrity through advanced MySQL features. By focusing on the context and prioritizing database-level reliability, the system provides a scalable foundation for real-world deployment.

Appendices

- Appendix A: Google Drive Link for the Full Project Folder

  https://drive.google.com/drive/folders/1jMus3ZmSdzu52IiEMQTFS8b9pGcRdAiG?usp=sharing